(MR22-1CS0110) DATA ANALYTICS

III Year B. Tech – II Semester

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COURSE OBJECTIVES:

- Learning about the Importance of Data and its importance
- Knowing Python fundamentals and Pandas essentials
- Learning the Principles of Probability and sampling Methods
- Getting knowledge about formulating and testing hypothesis
- Learning and analytical comparison's with ANOVA methods
- Learning about Performance indicators using ROC methods

UNIT-II

Introduction to Probability: Classical Probability, Relative Frequency, Sample Space, Events, Types of Probability, conditional Probability, Bayesian Rule, Relative frequency method, Random Variable, Distribution Function, Density Function

Sampling and Sampling Distribution: Random vs Non Random Sampling, Simple random sampling, cluster sampling, concept of sampling distributions, Student's t-test, Chi-square and F-distributions. Central limit theorem and its application, confidence intervals

Reference Books:

1. McKinney, W. (2012). Python for data analysis: Data wrangling with Pandas, NumPy, and IPython. " O'Reilly Media, Inc.".
2. Swaroop, C. H. (2003). A Byte of Python. Python Tutorial.
3. Ken Black, sixth Editing. Business Statistics for Contemporary Decision Making. “John Wiley & Sons, Inc”.
4. Anderson Sweeney Williams (2011). Statistics for Business and Economics. “Cengage Learning”.

Probability:

- ❖ Probability is a branch of mathematics that deals with the **likelihood or chance of different outcomes occurring in a particular event**.
- ❖ It provides a framework for quantifying uncertainty and randomness in various situations. There are different approaches to understanding and calculating probability, and **three fundamental concepts are classical probability, relative frequency, and Subjective Probability.**
- ❖ Probability is simply how likely something is to happen.
- ❖ Whenever we're unsure about the outcome of an event, we can talk about the probabilities of certain outcomes— **how likely they are**. The analysis of events governed by probability is called statistics.
- ❖ **A strong likelihood or chance of something.**
- ❖ **Probability is fundamental in statistics, machine learning, game theory, and everyday decision-making, making it an essential concept to understand.**

Terms Related to Probability

Experiment:

- ❖ An experiment is a type of action with unknown outcomes. There are a few positive outcomes and a few negative consequences in every experiment.
- ❖ Scientists will make thousands of unsuccessful attempts before they could make a successful attempt to make any invention.

Random Experiment:

- ❖ A random experiment is one in which the **set of possible outcomes is known**. Still, the specific outcome in a given experiment cannot be predicted before the experiment is carried out.
- ❖ **Example: Rolling a die, tossing a coin**

Trial:

- ❖ Trials are the various tries made during an experiment. In other words, a trial is any particular outcome of a random experiment.
- ❖ An event in probability is a specific outcome or a set of outcomes of an experiment. It is a subset of the sample space (all possible outcomes).
- ❖ **Example: Tossing a coin - An event with one outcome (e.g., rolling a "3" on a die).**

- ❖ **Event:**
- ❖ A trial with a clearly defined outcome is an event. For example, getting a tail when tossing a coin is termed an event.
- ❖ An event in probability is a specific outcome or a set of outcomes of an experiment. It is a subset of the sample space (all possible outcomes).
- ❖ **Example: An event with one outcome (e.g., rolling a "3" on a die).**

- **Random Event:**
- A random event cannot be easily foreseen. The chance value for such situations is extremely low. The appearance of a rainbow in the rain is a completely random occurrence.
- **Weather Forecast: Experiment: Observing the weather tomorrow.**
- **Random Event: It rains tomorrow or it does not rain.**
- **The occurrence of rain depends on multiple factors and is uncertain.**
- **Outcome:**
- The outcome of an event is a collection of all possible outcomes.
- An outcome is a single possible result of a random experiment. It represents an individual element of the sample space, which is the set of all possible outcomes of the experiment
- **Weather Forecast:**
- **Experiment: Observe the weather tomorrow.**
- **Possible Outcomes: {Rain, No Rain}{Rain, No Rain}.**
- **Example Outcome: Rain.**

Sample Space in Probability

- The sample space (denoted as S) in probability is the set of all possible outcomes of a random experiment.
- It represents every outcome that could potentially occur when the experiment is conducted.
- The sample space is the foundation of probability theory, listing all possible outcomes of an experiment.
- Understanding the sample space is crucial for defining events and calculating probabilities.
- **Types of Sample Spaces**
- **Finite Sample Space:**
- Contains a limited number of outcomes.
- Example: Rolling a 6-sided die ($S=\{1,2,3,4,5,6\}$S={1,2,3,4,5,6}).
- **Infinite Sample Space:**
- Contains an unlimited number of outcomes.
- Example: Measuring the exact time it takes for a car to complete a lap

  (S={all positive real numbers}).

Types of Probability.

- Probability can be categorized into various types, each with its own approach to defining and calculating the likelihood of events. The three main types of probability are Classical Probability, Empirical (or Relative Frequency) Probability, and Subjective Probability.

1. Classical Probability:

- This type of probability is based on a priori knowledge and assumes that all outcomes in the sample space are equally likely.
- It is most applicable to situations where each outcome is equally likely, such as flipping a fair coin or rolling a fair die.
- The classical probability of an event A is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes.

$$P(A) = \frac{\text{Number of Favorable Outcomes for Event A}}{\text{Total Number of Possible Outcomes}}$$

- **For example, when rolling a fair six-sided die, the probability of getting a 4 is 1/6 because there is only one favorable outcome (rolling a 4) out of the six possible outcomes (rolling a 1, 2, 3, 4, 5, or 6).**

2. Empirical (Relative Frequency) Probability:

- Empirical probability is based on observed frequencies from past events or experiments.
- It involves conducting experiments or observations and calculating the ratio of the number of times an event occurs to the total number of trials.
- As the number of trials increases, the relative frequency converges to the actual probability of the event.

$$P(A) = \frac{\text{Number of Times Event A Occurs}}{\text{Total Number of Trials}}$$

- For example, if you toss a coin 100 times and it comes up heads 60 times, the relative frequency of getting heads is 60/100 or 0.6.

3. Subjective Probability:

- Subjective probability is based on an individual's personal judgment or belief about the likelihood of an event occurring.
- It is subjective in nature and may vary from person to person, depending on their knowledge, experience, and perception.
- Subjective probabilities are often used in decision-making and situations where objective data is unusual.
- There is no specific formula for subjective probability; individuals assign probabilities based on their perception, experience, or other subjective factors.
- Subjective probability is a type of probability derived from an individual's personal judgment or opinion about whether a particular event will occur.
- Unlike classical or frequentist probability, which relies on mathematical calculations or empirical data, subjective probability reflects personal beliefs, experiences, and intuition.

Conditional Probability:

- Conditional probability is one of the types of probability in probability theory, where the probability of one event is dependent on the other event already happened.
- As this type of event is very common in real life, conditional probability is often used to determine the probability of such cases.
- It is denoted by $P(A|B)$, where A and B are events, and it reads as "the probability of A given B.“
- The formula for conditional probability is defined as:
  - $\boldsymbol{P(A|B)=P(A\cap B)\ /\ P(B)}$
- $P(A|B)$ is the probability of event A happening, given that event B has already happened.
- $P(A\cap B)$ is the probability of both events A and B happening.
- $P(B)$ is the probability of event B happening

Bayesian Rule

- ❖ Bayesian Rule, also known as Bayes' Theorem or Bayes' Rule, is a fundamental concept in probability theory.
- ❖ It provides a way to update the probability of a hypothesis based on new evidence or information. In data analytics, Bayes' Theorem is often used for statistical inference, machine learning, and decision-making.
- ❖ The formula for Bayes' Theorem is expressed as follows:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Where:

- ❑ P(A|B) is the probability of hypothesis A given the evidence B (posterior probability).
- ❑ P(B|A) is the probability of evidence B given the hypothesis A (likelihood).
- ❑ P(A) is the prior probability of hypothesis A (prior probability).
- ❑ P(B) is the probability of evidence B occurring (normalizing constant).

Bayes' Theorem has numerous applications in various fields. Here are a few examples:

Medical Diagnosis:

- ❖ Scenario: Suppose you have a rare medical condition (A) that affects only 1% of the population. There is a diagnostic test (B) for this condition, but it is not perfect and produces false positives.
- ❖ Application: Bayes' Theorem can be used to calculate the probability that you have the condition given a positive test result, taking into account the test's sensitivity and specificity.

Spam Filtering:

- ❖ Scenario: You receive an email, and your email system uses a spam filter. The filter looks at certain characteristics of emails (features) to determine whether an email is spam (A) or not.
- ❖ Application: Bayes' Theorem can be applied to update the probability of an email being spam based on observed features, improving the accuracy of the spam filter over time.

Quality Control:

- ❖ Scenario: A factory produces items, and there is a quality control test (B) to check whether an item is defective (A) or not.
- ❖ Application: Bayes' Theorem can help in updating the probability of an item being defective given a failed quality control test, incorporating information about the overall defect rate and the test's accuracy.

Legal Decision Making:

- ❖ Scenario: In a court trial, evidence (B) is presented, and the goal is to determine the guilt or innocence of the accused (A).
- ❖ Application: Bayes' Theorem can be used to update the probability of guilt or innocence based on the presented evidence, considering the likelihood of the evidence under both scenarios.

Customer Relationship Management (CRM):

- ❖ Scenario: A company wants to predict the likelihood of a customer (A) making a purchase based on their previous behavior and interactions (B).
- ❖ Application: Bayes' Theorem is used to update the probability of a customer making a purchase given recent interactions, helping tailor marketing strategies.

Weather Forecasting:

- ❖ Scenario: Meteorologists use historical weather data (A) and current observations to predict future weather conditions (B).
- ❖ Application: Bayes' Theorem can help in adjusting the probability of certain weather events based on new observations, providing more accurate and updated weather forecasts.

Machine Learning:

- ❖ Scenario: In a machine learning classification problem, you want to predict whether an image contains a specific object (A) based on features extracted from the image (B).
- ❖ Application: Bayes' Theorem is used in Bayesian classifiers to update the probability of an image containing the object given observed features.

Traffic Flow Prediction:

- ❖ Scenario: Transportation planners want to predict the probability of traffic congestion (A) based on historical traffic data and current conditions (B).
- ❖ Application: Bayes' Theorem helps in updating the probability of traffic congestion given recent observations, improving traffic management strategies.

Relative frequency method:

- ❖ The relative frequency method is a statistical approach used in data analytics to analyze and summarize data based on the observed frequencies or proportions of different outcomes.
- ❖ It involves calculating the relative frequency of each event, which is the proportion of times that event occurs relative to the total number of observations.
- ❖ This method is particularly useful for exploring and describing the distribution of categorical or discrete data.
- ❖ Here are the key steps involved in the relative frequency method:

Collect Data:

- ❖ Gather the relevant data, typically categorical or discrete, where observations fall into distinct categories or classes.

Count Frequencies:

- ❖ Count the number of occurrences of each category in the dataset. This creates a frequency distribution, showing how often each category appears.

Calculate Relative Frequencies:

- Calculate the relative frequency for each category by dividing the frequency of that category by the total number of observations. Mathematically, it can be expressed as:

$$\text{Relative Frequency of Category} = \frac{\text{Frequency of Category}}{\text{Total Number of Observations}}$$

Express as Percentages:

- Optionally, the relative frequencies can be expressed as percentages by multiplying them by 100.
- This helps in providing a more natural understanding of the distribution.

Visualize the Distribution:

- Represent the relative frequencies graphically using charts or graphs such as bar charts, pie charts, or histograms.
- Visualization aids in better understanding the patterns and trends in the data.

❖ Let's consider a simple example where you have collected data on the favorite colors of 100 people:

Blue: 30 people

Red: 20 people

Green: 15 people

Yellow: 10 people

Other: 25 people

Calculations:

$$\text{Relative Frequency of Blue} = \frac{30}{100} = 0.30$$

$$\text{Relative Frequency of Red} = \frac{20}{100} = 0.20$$

$$\text{Relative Frequency of Green} = \frac{15}{100} = 0.15$$

$$\text{Relative Frequency of Yellow} = \frac{10}{100} = 0.10$$

$$\text{Relative Frequency of Other} = \frac{25}{100} = 0.25$$

Random Variable

- A random variable is a mathematical concept used in probability theory and statistics to describe numerical outcomes that result from a random experiment, process, or phenomenon. In simpler terms, it is a variable whose values are determined by chance. Random variables are a key component in the study of probability and are used to model uncertain situations.
- There are two main types of random variables:

Discrete Random Variable:

- A discrete random variable is one that takes on a countable number of distinct values. These values are often isolated points on the number line.
- Examples of discrete random variables include the number of heads obtained when flipping a coin, the number of cars passing through a toll booth in an hour, or the number of emails received in a day.
- The probability distribution of a discrete random variable is often described using a probability mass function (PMF), which gives the probability of each possible value of the random variable.

Continuous Random Variable:

- A continuous random variable is one that can take on any value within a specified range or interval.
- Continuous random variables are associated with continuous phenomena and have an infinite number of possible values.
- Examples include the height of a person, the temperature in a room, or the time it takes for a computer to process a task.
- The probability distribution of a continuous random variable is described using a probability density function (PDF).
- Unlike the PMF for discrete random variables, the PDF does not directly give probabilities for specific values but instead provides the probability density over intervals.

The Distribution Function

- In the theoretical discussion on Random Variables and Probability, we note that the probability distribution encouraged by a random variable X is determined uniquely by a consistent assignment of mass to semi-infinite intervals of the form $(-\infty, t]$ for each real t.
- This suggests that a natural description is provided by the following.
- **Definition**
- The distribution function F_X for random variable X is given by

$$F_X(t) P(X \leq t) = P(X \in (-\infty, t]) \; \forall t \in R$$

- In terms of the mass distribution on the line, this is the probability mass at or to the left of the point t. As a consequence

Density Function

- A density function, also known as a probability density function (PDF), is a function that describes the likelihood of a random variable appearing as a certain value.
- The function's value at any given sample in the sample space can be interpreted as the relative likelihood that the value of the random variable would be equal to that sample

Sample Definition

- The sample or respondents for this study may be chosen from a group of **known or unknowing consumers or customers.**
- Often times, even though you are aware of the typical respondent profile, you cannot complete your research project without the respondents.
- In these circumstances, researchers and research teams get in touch with specialized organizations to use their respondent panel or purchase respondents from them to finish research studies and surveys.
- They could be respondents from the general population who meet the demographic requirements or those who meet certain criteria.
- The success of research investigations depends on these responders. The many sample types, sampling techniques, and representative instances are covered in length in this page. Additionally, it describes how to compute the size, provides information about an online sample, and highlights the benefits of employing them.

What is a Sample?

- A sample is a summarized set of information that a **researcher selects or picks from a broader population using a predetermined technique of selection**. These components are referred to as **observations, sampling units, or sample points.**
- Developing a sample is a productive way to carry out research. The entire population must frequently be studied, which is difficult, expensive, and time consuming.
- As a result, studying the sample offers information the researcher can use to understand the complete population.
- For instance, a cell phone manufacturer might want to interview students at American universities about certain features.
- If the researcher wants to find features that students utilize, features they would want to see, and the price they are prepared to pay, an extensive research study must be carried out.
- It is crucial to complete this step in order to comprehend the features that need to be developed, those that need to be upgraded, the device's price, and the go-to-market plan.

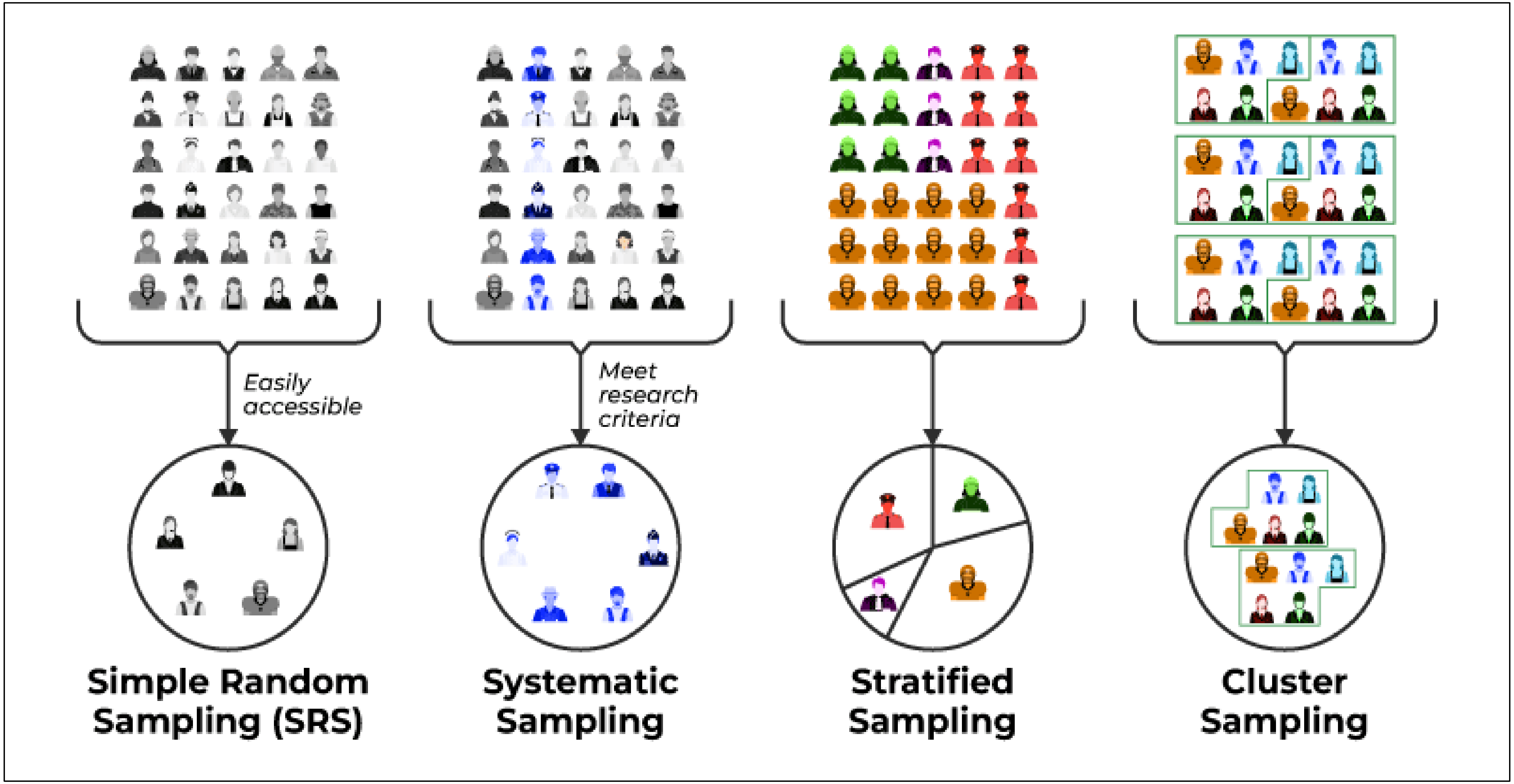
- There were 24.7 million students enrolled in American universities in 2016-17 alone.
- It is impossible to investigate all of these pupils; the time and money required to do so would render the study meaningless and the new technology unnecessary.
- A sufficient sample of students for study can be obtained by selecting universities based on their geographic location and then selecting a subset of their students.
- The population for market research is typically rather large. The entire population cannot be counted, in all likelihood.
- Typically, the sample represents a sizeable portion of this population.
- Surveys, polls, and questionnaires are then used by researchers to gather data from these samples, and this data analysis is extrapolated to the larger community.

Types of Samples: selection methodologies with examples

- A sampling method is the procedure used to get a sample.
- While this technique generates the quantitative and qualitative data that can be collected as part of a research study, sampling plays a crucial role in the research design.
- Probability sampling and non-probability sampling are two separate approaches to sampling techniques.

PROBABILITY SAMPLING METHODOLOGIES

- The process of obtaining a sample through probability sampling involves choosing the objects from a population according to probability theory.
- Everyone in the population is included in this technique, and everyone has an equal chance of getting chosen. Hence, there is absolutely no bias in this kind of sample. The research can then involve every member of the population.
- The selection criteria are chosen at the beginning of the market research study and are a crucial part of the investigation.
- **Four different types of samples can be used in probability sampling. They are:**


Easily accessible
Meet research criteria
Simple Random Sampling (SRS)
Systematic Sampling
Stratified Sampling
Cluster Sampling

1. Simple random sampling

- This method of sample selection is the easiest to understand. Each participant has an equal chance of taking part in the study using this strategy. Each member of this sample population has an equal chance of being chosen at random to become an object.
- It is one of the most basic and commonly used forms of probability sampling.
- For Example, if a university dean wanted to gather input from students about how they felt about the professors and their level of education, this sample could include all 1000 students at the university. To create this sample, 100 students can be chosen at random from any class.

Steps in Simple Random Sampling

- **Define the Population:**
  - Clearly identify the population you want to study.
  - Example: All 1,000 students in a school.
- Assign Numbers to Each Member:
  - Assign a unique identifier (e.g., student ID or a sequential number) to every individual in the population.

- **Select the Sample Randomly:**
    - Use a random selection method, such as:
    - Random number generators
    - Lottery methods (drawing names or numbers from a hat)
- **Collect Data from the Selected Sample:**
    - Only the randomly selected individuals are surveyed or analyzed.

Advantages of Simple Random Sampling

- **Unbiased:** Every individual has an equal chance of selection, reducing selection bias.
- **Easy to Implement:** Simple and straightforward with modern tools (e.g., random number generators).

Disadvantages of Simple Random Sampling

- **Not Always Practical:** It may be challenging to list every individual in a large or dispersed population.
- **May Not Guarantee Representation:** Random selection could lead to an unbalanced sample (e.g., more males than females).

2. Cluster sampling

- A sample technique called cluster sampling divides the respondent population into equal clusters.
- Cluster sampling is a sampling method where the population is divided into groups, called clusters, and then entire clusters are randomly selected to collect data from all members of the chosen clusters. This is especially useful when the population is large and geographically disseminated.
- **Steps in Cluster Sampling**
- **Divide the Population into Clusters:**
  - Divide the entire population into non-overlapping, naturally occurring groups or clusters (e.g., schools, neighborhoods, or districts).
  - Clusters should ideally represent the diversity of the population
- **Select Clusters Randomly:**
  - Use a random sampling method to select a few clusters. You do not need to select every cluster.
- **Collect Data:**
  - Data is collected from all individuals within the selected clusters

3. Systematic sampling

- By using systematic sampling, a population is sampled by randomly selecting respondents at equal intervals. Picking a beginning point and then choosing responders at a predetermined sample interval is the method for choosing the sample.
- Systematic Sampling is a probability sampling method where you select every kk-th individual from a population list, starting from a randomly chosen starting point. This method is simpler and quicker than simple random sampling, especially for large populations.
- As an diagram, when choosing 1,000 volunteers for the Olympics from a list of 10,000 applicants, each applicant is assigned a count between 1 and 10,000. Then a sample of 1,000 volunteers can be obtained by counting backwards from 1 and selecting each respondent with a 10-second interval.

Advantages of Systematic Sampling

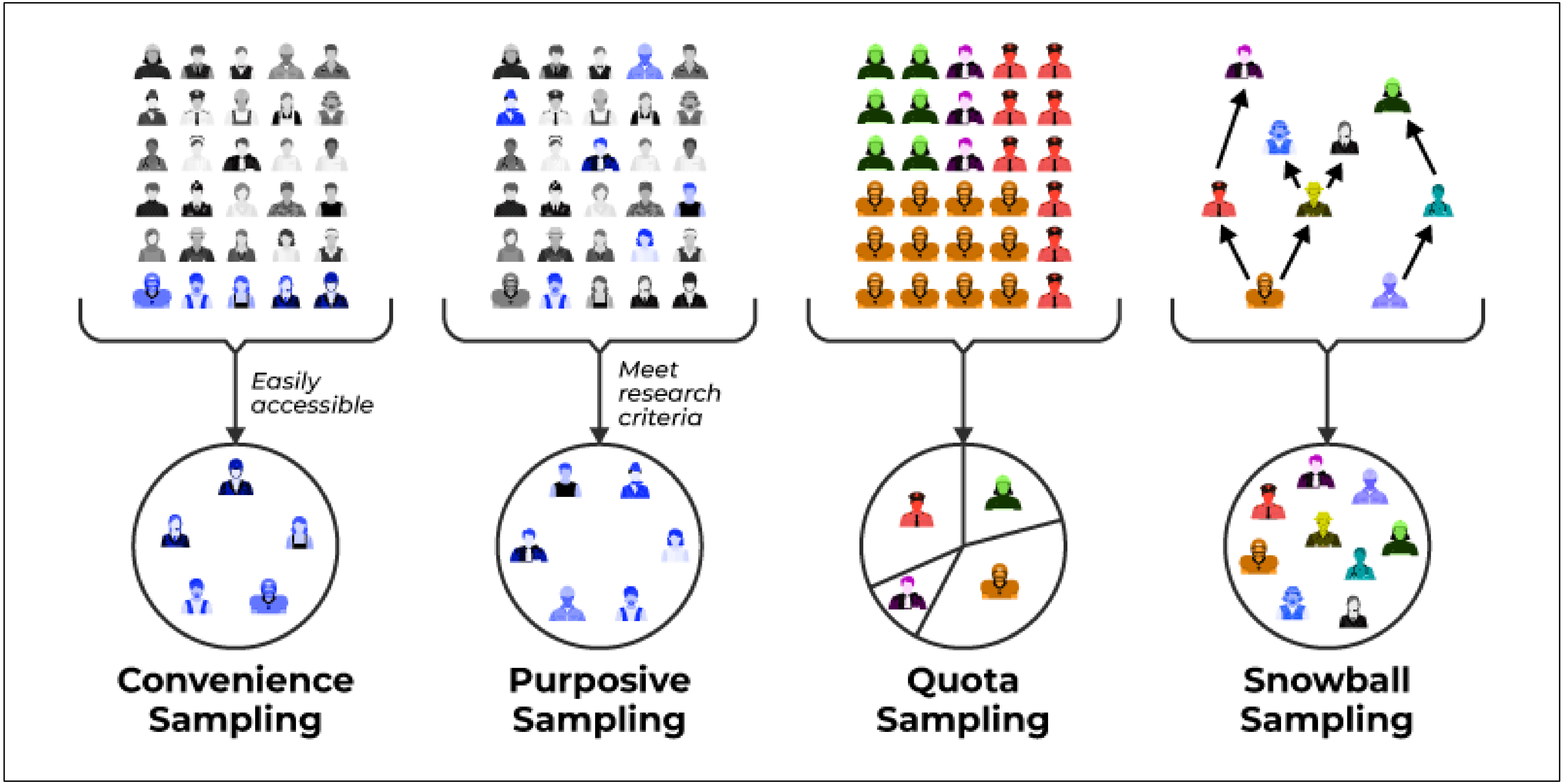
- Simplicity: Easier and faster to implement than simple random sampling.
- Efficiency: Works well with ordered populations, such as lists or production lines.
- Ensures Even Coverage: By spreading the sample evenly across the population, systematic sampling can prevent clustering.

4. Stratified random sampling

- In the research design phase, stratified random sampling is a technique for segmenting the respondent population into discrete but pre-defined parameters. The responders in this method don't overlap; rather, they speak for the entire population as a whole.
- Stratified random sampling is a probability sampling method where the population is divided into distinct subgroups, called strata, based on shared characteristics.
- A random sample is then taken from each stratum to ensure that the sample is representative of the population as a whole
- For Example, a researcher examining persons from various socioeconomic backgrounds can identify respondents based on their yearly earnings. Afterward, some of the objects from these samples can be employed for the research study. This creates smaller groups of persons or samples.

Non-probability sampling methodologies with examples

- ❖ Non-probability sampling refers to sampling techniques where not all members of the population have an equal chance of being selected. These methods are often used when random sampling is not feasible due to time, cost, or population constraints. While non-probability sampling is less rigorous than probability sampling, it is widely used in exploratory research and practical applications.
- ❖ The researcher's judgment is used to choose a sample in the non-probability sampling technique.
- ❖ This kind of sample is primarily determined by the researcher's or mathematician's capacity to access it.
- ❖ When conducting preliminary research, this kind of sampling is employed since the main goal is to generate a hypothesis regarding the research issue.
- ❖ Here, each participant does not have an equal probability of being in the sample population, and the sample is only made aware of these parameters after it has been chosen.
- ❖ Non-probability sampling can be divided into **four different kinds of samples**. They are:


Easily accessible
Meet research criteria
Convenience Sampling
Purposive Sampling
Quota Sampling
Snowball Sampling

1. Convenience sampling

- In plain English, convenience sampling refers to the ease with which a researcher can contact a respondent.
- The method used to create this sample is not scientific.
- Researchers collect data from readily available participants without random selection.
- The selection of the sample components is done only on the basis of proximity, not representativeness, and the researchers have almost no control over it.
- When there are time and financial constraints on gathering feedback, this non-probability sampling method is used.
- As an Example, consider researchers who are conducting a mall-intercept study to determine the likelihood that people will use a **perfume** produced by a perfume business.
- Based on their closeness to the survey desk and desire to engage in the study, the sample respondents in this sampling technique are selected.

2. Judgmental/purposive sampling

- The judgmental or purposive sampling approach is a way of selecting a sample based only on the researcher's judgment and understanding of the target audience, the nature of the study, and other relevant factors. Only those individuals who meet the research criteria and end goals are chosen using this sample technique, while the rest are excluded. If the research question is "Would you like to do your Masters?" and the only acceptable response is "Yes," then everyone else is not included in the study. As an illustration, if the research question is "What University do you prefer as a student for Masters?"

3. Snowball sampling

- A non-probability sampling method where the samples have uncommon characteristics is known as **snowball sampling or chain-referral sampling**. This sampling method uses recommendations from current participants to find the sample populations needed for a study. For instance, when asked for feedback on a sensitive subject like AIDS, respondents are quiet to provide details. In this situation, the researcher can enlist individuals who have expertise or understanding of such individuals and ask them to gather information on behalf of the researcher.

4. Quota sampling

- With quota sampling, the researcher is free to choose the sample they want to use based on their stratification. This method's main characteristic is that two persons cannot coexist in two different environments.
- For instance, a shoe producer could want to comprehend how millennial view the brand in relation to other factors like comfort, cost, etc. For this study, it solely chooses female millennial because the goal is to gather opinions on women's shoes.

Advantages and Disadvantages : Non-Probability Sampling and Probability Sampling

Sampling Type	Advantages	Disadvantages
Non-Probability Sampling	❖ Easy to execute and cost-effective. ❖ Suitable for preliminary or exploratory research.	❖ High bias risk. ❖ Results may not represent the population accurately.
Probability Sampling	❖ Accurate representation of the population. ❖ Enables statistical inferences and generalizations.	❖ Time-consuming and expensive. ❖ Requires a complete population list.

Understanding T-values and P-values

- The P-value and T-value are important concepts in statistical hypothesis testing. They are used to assess whether the results of an analysis are statistically significant.
- T-Value (Test Statistic):
- The T-value is a calculated value that measures the difference between the observed data and what is expected under the null hypothesis, expressed in terms of standard errors.
- It is part of the t-test, which is used to compare the means of groups or to assess the significance of a coefficient in a regression model.
- **Formula for T-value, For a one-sample t-test:**

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Where:

- $\bar{x}$: Sample mean
- μ: Population mean under the null hypothesis
- s: Standard deviation of the sample
- n: Sample size

Understanding:

- ❖ A large absolute T-value indicates a large difference between the observed data and the null hypothesis.
- ❖ The sign of the T-value indicates the direction of the difference.

P-Value

- ❖ The P-value is the probability of observing a test statistic (like the T-value) as extreme as, or more extreme than, the observed value, assuming the null hypothesis is true.
- ❖ **Key Points about P-Value:**
- ❖ **Range:** The P-value is between 0 and 1.
- ❖ Threshold for significance: A common threshold is 0.05 (5%).
  - ❖ If $P \leq 0.05$: Reject the null hypothesis (results are statistically significant).
  - ❖ If $P > 0.05$: Fail to reject the null hypothesis (results are not statistically significant).
- ❖ Example: If the P-value is 0.03, it means there is a 3% probability that the observed data could occur by random chance under the null hypothesis.

Relationship Between T-Value and P-Value

- ❖ T-value helps determine how far the sample statistic (e.g., sample mean) is from the null hypothesis.
- ❖ The P-value is derived from the T-value using the t-distribution. It quantifies the evidence against the null hypothesis.
- ❖ Larger absolute T-values usually correspond to smaller P-values.

T-Value (Absolute)	**P-Value** (Probability)	**Understanding**
Very Small (Close to 0)	Close to 1	Weak evidence against the null hypothesis (not significant).
Moderate	Around 0.05 (or slightly more)	Borderline significance. May or may not reject the null.
Large	Less than 0.05	Strong evidence against the null hypothesis (significant).
Very Large	Close to 0	Extremely strong evidence against the null hypothesis.

CHI-SQUARE TEST

- A chi-square test is a statistical test that is used to **compare observed and expected results**.
- A chi-square test is a statistical hypothesis test that surveys whether two categorical variables are independent in influencing the test statistic.
- It's used to compare observed results with expected results to determine if a difference is due to chance or a relationship between the variables.
- **The goal of this test is to identify whether a inequality between actual and predicted data is due to chance or to a link between the variables under consideration**.
- A chi-square test or comparable nonparametric test is required to test a hypothesis regarding the distribution of a categorical variable.
- Categorical variables, which indicate categories such as animals or countries, can be nominal or ordinal. They cannot have a normal distribution since they can only have a few particular values.

- It is used to calculate the difference between two categorical variables, which are:
    - As a result of chance
    - Because of the relationship
- Formula For Chi-Square Test

$$x_c^2 = \frac{\Sigma\,(O_i - E_i)^2}{E_i}$$

Where

c = Degrees of freedom

O = Observed Value

E = Expected Value

- The degrees of freedom in a statistical calculation represent the number of variables that can vary in a calculation. The degrees of freedom can be calculated to ensure that chi-square tests are statistically valid. These tests are frequently used to compare observed data with data that would be expected to be obtained if a particular hypothesis were true.
- The Observed values are those you gather yourselves.
- The expected values are the frequencies expected, based on the null hypothesis.

Use of Chi-Square Test:

- Here are some of the uses of the Chi-Squared test:
- The Chi-squared test can be used to see if your data follows a well-known theoretical probability distribution like the Normal or Poisson distribution.
- The Chi-squared test allows you to assess your trained regression model's goodness of fit on the training, validation, and test data sets.
- **Karl Pearson introduced this test in 1900 for categorical data analysis and distribution. This test is also known as 'Pearson's Chi-Squared Test'.**
- Chi-Squared Tests are most commonly used in hypothesis testing. A hypothesis is an assumption that any given condition might be true, which can be tested afterwards.
- The Chi-Square test estimates the size of inconsistency between the expected results and the actual results when the size of the sample and the number of variables in the relationship is mentioned.

- These tests use degrees of freedom to determine if a particular null hypothesis can be rejected based on the total number of observations made in the experiments. Larger the sample size, more reliable is the result.
- There are two main types of Chi-Square tests namely –
  - Independence
  - Goodness-of-Fit

Independence

- The Chi-Square Test of Independence is a derivable statistical test which observes whether the two sets of variables are likely to be related with each other or not.
- This test is used when we have counts of values for two nominal or categorical variables and is considered as non-parametric test.
- A relatively large sample size and independence of observations are the required criteria for conducting this test.

For Example-

- In a movie theatre, suppose we made a list of movie genres.
- Let us consider this as the first variable. The second variable is whether or not the people who came to watch those genres of movies have bought snacks at the theatre.
- Here the null hypothesis is that the genre of the film and whether people bought snacks or not are unreliable.
- If this is true, the movie genres don’t impact snack sales.

Goodness-Of-Fit

- In statistical hypothesis testing, the Chi-Square Goodness-of-Fit test determines whether a variable is likely to come from a given distribution or not.
- To test whether an observed distribution matches an expected distribution. We can use this test when we have value counts for categorical variables.
- This test demonstrates a way of deciding if the data values have a “ good enough” fit for our idea or if it is a representative sample data of the entire population.

Scenario

A company wants to analyze the relationship between employees' education level and their job satisfaction. The data is presented in a contingency table that categorizes employees based on their education levels (High School, Bachelor's, and Master's) and their job satisfaction levels (Satisfied, Neutral, and Dissatisfied). Using the Chi-Square Test for Independence, determine whether there is a statistically significant relationship between education level and job satisfaction.

Education Level	Satisfied	Neutral	Dissatisfied
High School	50	30	20
Bachelor's	40	25	35
Master's	30	20	50

Scenario

Testing Relationship Between Education Level and Job SatisfactionWe have a contingency table showing the counts of employees' satisfaction levels (Satisfied, Neutral, Dissatisfied) across three education levels (High School, Bachelor's, Master's)

Education Level	Satisfied	Neutral	Dissatisfied
High School	50	30	20
Bachelor's	40	25	35
Master's	30	20	50

Import Required Libraries

```
1. import numpy as np
2. from scipy.stats import chi2_contingency
```

❖ chi2_contingency: This function from the scipy.stats library is used to perform the Chi-Square Test for Independence.

Define the Contingency Table

```
1. data = [
2.     [50, 30, 20],  # High School
3.     [40, 25, 35],  # Bachelor's
4.     [30, 20, 50]   # Master's
5. ]
```

❖ The data represents the observed frequencies.

❖ Rows represent categories of one variable (Education Level: High School, Bachelor's, Master's).

❖ Columns represent categories of another variable (Job Satisfaction: Satisfied, Neutral, Dissatisfied).

Perform the Chi-Square Test

1. chi2, p, dof, expected = chi2_contingency(data)

- chi2_contingency(data):
- Takes the contingency table as input.
- Returns:
- **Chi-Square statistic (chi2):** Measures the difference between observed and expected frequencies.
- **p-value (p):** Probability of observing the data assuming the null hypothesis is true.
- **Degrees of Freedom (dof):** Calculated as $(rows-1)\times(columns-1)$
- **Expected frequencies (expected):** Frequencies expected if the variables are independent

Display Results

1. print("Chi-Square Test Results:")
2. print(f"Chi-Square Statistic: {chi2:.4f}")
3. print(f"Degrees of Freedom: {dof}")
4. print(f"P-Value: {p:.4f}")
5. print("\nExpected Frequencies:")
6. print(expected)

❖ Outputs the Chi-Square statistic, degrees of freedom, p-value, and the expected frequencies.

Interpret the Results

```
alpha = 0.05  # Significance level
if p < alpha:
    print("\nResult: Reject the null hypothesis (variables are dependent).")
else:
    print("\nResult: Fail to reject the null hypothesis (variables are independent).")
```

- ❖ Significance level (alpha): Usually set at 0.05.
- ❖ Hypotheses
- ❖ Null Hypothesis (H_0): The two variables are independent (no relationship).
- ❖ Alternative Hypothesis (H_1): The two variables are dependent (there is a relationship).

Scenario

Testing Relationship Between Education Level and Job SatisfactionWe have a contingency table showing the counts of employees' satisfaction levels (Satisfied, Neutral, Dissatisfied) across three education levels (High School, Bachelor's, Master's)

Education Level	Satisfied	Neutral	Dissatisfied
High School	50	30	20
Bachelor's	40	25	35
Master's	30	20	50

For mathematically calucution:

Calculate the Chi-Square Statistic

❖ The Chi-Square statistic ($\chi 2$) is computed using the formula:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where:

❖ O: Observed frequency (from the input table).

❖ E: Expected frequency (calculated assuming the variables are independent).

❖ From the given contingency table

Education Level	Satisfied	Neutral	Dissatisfied	Row Totals
High School	50	30	20	100
Bachelor's	40	25	35	100
Master's	30	20	50	100
Column Totals	120	75	105	**300**

The expected frequencies are calculated as:

$$E_{ij} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

For example: **Expected value for High School & Satisfied**:

$$E = \frac{100 \times 120}{300} = 40$$

Repeat this for all cells.

Expected value for High School & Neutral, High School & Dissatisfied

And

Expected value for Bachelor's & Satisfied, Bachelor's & Neutral, Bachelor's & Dissatisfied

Expected value for Master's & Satisfied, Master's & Neutral, Master's & Dissatisfied

First Row (High School)

- $E_{11} = \frac{(100 \times 120)}{300} = 40.00$
- $E_{12} = \frac{(100 \times 75)}{300} = 25.00$
- $E_{13} = \frac{(100 \times 105)}{300} = 35.00$

Second Row (Bachelor's)

- $E_{21} = \frac{(100 \times 120)}{300} = 40.00$
- $E_{22} = \frac{(100 \times 75)}{300} = 25.00$
- $E_{23} = \frac{(100 \times 105)}{300} = 35.00$

Third Row (Master's)

- $E_{31} = \frac{(100 \times 120)}{300} = 40.00$
- $E_{32} = \frac{(100 \times 75)}{300} = 25.00$
- $E_{33} = \frac{(100 \times 105)}{300} = 35.00$

The expected table is:

Education Level	Satisfied	Neutral	Dissatisfied
High School	40.00	25.00	35.00
Bachelor's	40.00	25.00	35.00
Master's	40.00	25.00	35.00

Compute Chi-Square Statistic ($\chi 2$):

The Chi-Square statistic is computed as: $$\chi^2 = \sum \frac{(O-E)^2}{E}$$

For each cell:

First Row (High School)

- $\frac{(50-40)^2}{40} = \frac{(10)^2}{40} = \frac{100}{40} = 2.50$
- $\frac{(30-25)^2}{25} = \frac{(5)^2}{25} = \frac{25}{25} = 1.00$
- $\frac{(20-35)^2}{35} = \frac{(-15)^2}{35} = \frac{225}{35} = 6.43$

Second Row (Bachelor's)

- $\frac{(40-40)^2}{40} = \frac{(0)^2}{40} = 0.00$
- $\frac{(25-25)^2}{25} = \frac{(0)^2}{25} = 0.00$
- $\frac{(35-35)^2}{35} = \frac{(0)^2}{35} = 0.00$

Third Row (Master's)

- $\frac{(30-40)^2}{40} = \frac{(-10)^2}{40} = \frac{100}{40} = 2.50$
- $\frac{(20-25)^2}{25} = \frac{(-5)^2}{25} = \frac{25}{25} = 1.00$
- $\frac{(50-35)^2}{35} = \frac{(15)^2}{35} = \frac{225}{35} = 6.43$

Summing all these values:

$$\chi^2 = 2.50 + 1.00 + 6.43 + 0.00 + 0.00 + 0.00 + 2.50 + 1.00 + 6.43 = \mathbf{19.86}$$

Thus, the Chi-Square statistic is: $$\chi^2 = \mathbf{19.86}$$

Compute Degrees of Freedom

The degrees of freedom (dof) is given by:

$$\text{dof} = (\text{Number of Rows} - 1) \times (\text{Number of Columns} - 1)$$

For this table:

$$\text{dof} = (3-1) \times (3-1) = 2 \times 2 = 4$$

Compute the P-Value

- ❖ The p-value is determined by finding the area under the Chi-Square distribution curve to the right of the calculated $\chi 2$ statistic (16.5714) with 4 degrees of freedom.
- ❖ This is calculated using the Chi-Square cumulative distribution function (CDF):

$$\text{p-value} = 1 - \text{CDF}(\chi 2, \text{dof})$$

- ❖ Using a statistical table or software:

$$\text{p-value} = 1 - \text{CDF}(19.86, 4) = 0.0054$$

```python
import scipy.stats as stats
# Parameters
x = 19.86 # Value
df = 4
# Degrees of freedom
#Compute the CDF
chi_square_cdf = stats.chi2.cdf(x, df)
chi_square_cdf
```

```
0.999467798668642
```

(df)	.99	.975	.95	.9	.1	.05	.025	.01
1	-------	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932

Decision:

- Since the p-value (0.0054) is less than the significance level (0.05), we reject the null hypothesis, concluding that education level and job satisfaction are related.

Student’s t-Test

- ❖ Student’s t-test is a statistical test used to compare the means of two groups to determine if they are significantly different from each other. It is commonly used when the sample size is small (typically less than 30) and the population standard deviation is unknown..
- ❖ We can define the Student t-test as a method that tells you how significant the differences can be between different groups.
- ❖ **A Student t-test is defined as a statistic and this is used to compare the means of two different populations.**
- ❖ It is a method that is often used in hypothesis testing to find out whether a process or whether a given treatment actually has any effect on the population of interest, or whether or not two populations are different from each other.
- ❖ You can test the difference between these two groups with the help of the Student t-test.
- ❖ The null hypothesis (H0) is one that tells the true difference between these groups.
- ❖ The alternate hypothesis (Ha) is one that tells the true difference is different from zero.

Student t Test Introduction

- In the year 1908, an Englishman named **William Sealy Gosset** developed the t-test as well as t distribution.
- **William** worked at the Guinness brewery in Dublin and found which existing statistical techniques using large samples were not useful for the small sample sizes which he encountered in his work.
- The **t distribution** belonging under a family of curves in which the number of degrees of freedom specifies a particular curve.

Purpose of Student's t-Test

- To compare the means of two groups and determine if they are significantly different.
- To test hypotheses about population means using sample data.
- To assess the impact of an intervention or treatment in experiments.

Student t-Test Formula

❖ The formula for the two-sample t-test / the Student's t-test) is shown below.

$$\text{Student t Test Formula, } t = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{(s^2(\frac{1}{n_1} + \frac{1}{n_2}))}}$$

❖ In the formula given above, **t is equal to the t-value**, **x1 and x2** are the means of the two groups being compared, **$s^2$is the pooled standard error** of the two groups, and **n1 and n2 are the numbers of observations in each of the groups**.

❖ A **larger t-value denotes the difference between group means is greater than the pooled standard error**, indicating a more significant difference between the groups.

❖ You can compare your calculated t-value against the values in a critical value chart to determine whether your t-value is greater than what would be expected by chance.

❖ If so, you can reject the null hypothesis and you can conclude which two groups are in fact different.

Types of Student t-Test

- When choosing a Student t-test, two things need to be kept in mind: whether the groups being compared are coming from a single population or two different populations, There are different types of t-tests, but the two most common ones are.
    1. **Independent Samples T-Test**
    2. **Paired Samples T-Test**

1. Independent Samples T-Test:

- This test is used when comparing the means of two independent groups. For example, comparing the average scores of two different groups of participants in an experiment or comparing the means of two different treatment groups.
- The formula for the t-statistic in the independent samples t-test is:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- Where X1 and X2 are the sample means s1 and s2 are the sample standard deviations, and n1 and n2 are the sample sizes.

Paired Samples T-Test:

- ❖ This test is used when comparing the means of two related groups. For example, comparing the scores of the same group of participants before and after a treatment.
- ❖ The formula for the t-statistic in the paired samples t-test is:

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$$

- ❖ where d is the mean of the differences between paired observations, sd is the standard deviation of the differences, and n is the number of pairs.

- **A company wants to evaluate the effectiveness of its training program on employee performance. To assess this, the performance scores of 8 employees were recorded both before and after the training. Using the Student's t-test, determine whether the training program has led to a statistically significant improvement in employee performance.**
- **Given Performance Scores:**
- **Before Training: [65, 70, 68, 72, 75, 78, 74, 71]**
- **After Training: [78, 82, 80, 85, 87, 90, 86, 83]**
- **Perform a paired t-test and interpret the results.**

- **A company seeks to evaluate the effectiveness of its training program on employee performance. To achieve this, performance scores of 8 employees are recorded both before and after the training. The objective is to determine whether the training program has led to a statistically significant improvement in employee performance.**

- This program evaluates whether a training program has significantly improved employee performance using a paired t-test. The paired t-test compares the means of two related groups (before and after training) to check for significant differences.

- Here Assume the Performance Scores
  - before_training = [65, 70, 68, 72, 75, 78, 74, 71]
  - after_training = [78, 82, 80, 85, 87, 90, 86, 83]

- Two lists represent performance scores before and after training for 8 employees.
- Each index in before_training corresponds to the same employee's score in after_training.

- **Import Required Libraries**

1. import numpy as np
2. from scipy import stats

- numpy is imported for numerical computations (though not explicitly used here).scipy.stats is imported to perform the paired t-test (ttest_rel function).

- **Define the Data (Performance Scores)**

1. before_training = [65, 70, 68, 72, 75, 78, 74, 71]
2. after_training = [78, 82, 80, 85, 87, 90, 86, 83]

- Two lists represent performance scores before and after training for 8 employees.
- Each index in before_training corresponds to the same employee’s score in after_training.

❖ **Perform the Paired t-Test**

```
1.   t_stat, p_value = stats.ttest_rel(before_training, after_training)
```

❖ ttest_rel() is used for paired data (same employees before and after training).

❖ This function returns:

❖ t_stat: The t-statistic, a measure of difference between means.

❖ p_value: The probability that the difference is due to random chance.

❖ **Print the t-Test Results**

```
1.   print(f"T-statistic: {t_stat:.3f}")
2.   print(f"P-value: {p_value:.3f}")
```

- **Interpret the Results**

```python
alpha = 0.05
if p_value < alpha:
    print("Reject the null hypothesis: The training significantly improved performance.")
else:
    print("Fail to reject the null hypothesis: No significant improvement due to training.")
```

- alpha = 0.05 (5% significance level) is a common threshold in hypothesis testing.
- If p_value < 0.05, we reject the null hypothesis, concluding significant improvement.
- Otherwise, we fail to reject the null hypothesis, meaning no strong evidence that training improved performance.

- **Mathematical Calculation for Paired t-Test**
- We will now manually compute the paired t-test step by step using the given data.
- **Given Data (Performance Scores)**

Employee	Before Training (Xi)	After Training (Yi)	Difference (Di = Yi−Xi)
1	65	78	13
2	70	82	12
3	68	80	12
4	72	85	13
5	75	87	12
6	78	90	12
7	74	86	12
8	71	83	12
Sum	**573**	**671**	**96**

❖ Calculate the Mean Difference

$$\bar{D} = \frac{\sum D_i}{n}$$

$$\bar{D} = \frac{96}{8} = 12$$

❖ Calculate the Standard Deviation of Differences

The formula for the standard deviation of the differences (s_D) is:

$$s_D = \sqrt{\frac{\sum (D_i - \bar{D})^2}{n-1}}$$

❖ Calculate $(D_i - \bar{D})^2$ for Each Employee

Employee	Di	(Di−D¯)	(Di−D¯)^2
1	13	13−12=1	1
2	12	12−12=0	0
3	12	12−12=0	0
4	13	13−12=1	1
5	12	12−12=0	0
6	12	12−12=0	0
7	12	12−12=0	0
8	12	12−12=0	0
Sum	**96**	**-**	**2**

$$s_D = \sqrt{\frac{2}{8-1}} = \sqrt{\frac{2}{7}} = \sqrt{0.2857} \approx 0.535$$

❖ Calculate the t-Statistic

The paired t-test formula is:

$$t = \frac{\bar{D}}{s_D/\sqrt{n}}$$

Substituting the values:

$$t = \frac{12}{\frac{0.535}{\sqrt{8}}}$$

$$t = \frac{12}{0.1892} \approx 63.44$$

- **Determine the p-value and Conclusion**
- From the t-table, for degrees of freedom (df = n - 1 = 8 - 1 = 7) at a significance level of 0.05, the critical t-value is 2.447.Since t = 63.44 is much greater than 2.447, the p-value is very small ($p<0.001$), leading to rejection of the null hypothesis.

A **t-table** provides critical values for different degrees of freedom and confidence levels. Below is an excerpt:

df	t (0.10, two-tailed)	t (0.05, two-tailed)	t (0.01, two-tailed)
6	1.943	**2.447**	3.707
7	1.895	**2.447**	3.499
8	1.860	**2.306**	3.355

Since our **df = 7** and $\alpha = 0.05$, we find **t = 2.447**.

- df=n−1=8−1=7Since we have 8 employees, the degrees of freedom (df) = 7.

- Compare t-Statistic with t-Critical
- t-Statistic from our calculation: 63.44
- t-Critical (from table): 2.447
- Since 63.44 > 2.447, we reject the null hypothesis and conclude a significant difference.

- Instead of using a t-table manually, we can compute the t-critical value using Python:

```
1. from scipy.stats import t
2. # Define degrees of freedom
3. df = 7
4. # Compute t-critical for a two-tailed test with α = 0.05
5. t_critical = t.ppf(1 - 0.05 / 2, df)
6. print(f"T-critical value: {t_critical:.3f}")
```

- **Output:**
- **T-critical value: 2.447**

Central limit theorem and its application, confidence intervals

- Central Limit Theorem (CLT)The Central Limit Theorem (CLT) states that, regardless of the shape of the population distribution, the sampling distribution of the sample mean will be approximately normal if the sample size is sufficiently large (typically n ≥30n≥30).
- **Key points about the CLT:**
- **Large sample size:** The CLT is most accurate when the sample size is large, typically considered to be 30 or more samples.
- **Independent samples:** The samples taken must be independent of each other.
- **Population mean and variance:** The CLT allows us to estimate the population mean and variance using the sample mean and variance, even if the original population distribution is not normal.

Central Limit Theorem Formula

- Let us assume we have a random variable X. Let σ be its standard deviation and μ is the mean of the random variable.
- Now as per the Central Limit Theorem, the sample mean X‾ will approximate to the normal distribution which is given as X‾ ~ N(μ, σ/√n).
- The Z-Score of the random variable X‾ is given as Z . Here x‾ is the mean of X‾. The image of the formula is attached below.

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Sample Mean = Population Mean = μ

$$\text{Sample Standard Deviation} = \frac{\text{Standard Deviation}}{n}$$

OR

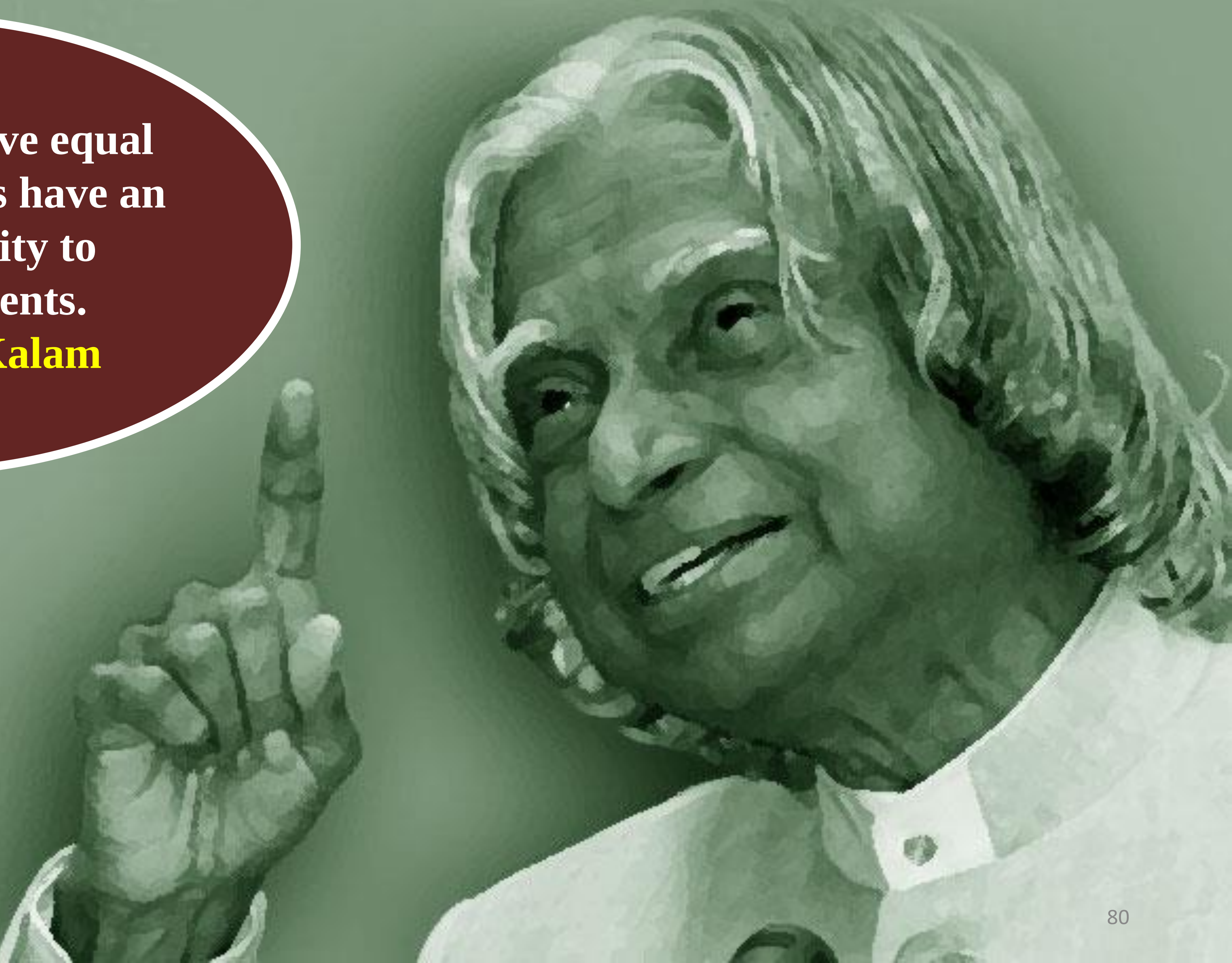
$$\text{Sample Standard Deviation} = \frac{\sigma}{\sqrt{n}}$$

Applications of the Central Limit Theorem (CLT)

- ❖ **Hypothesis Testing** – Ensures sample means follow a normal distribution for statistical tests.
- ❖ **Confidence Intervals** – Helps estimate population parameters with a range of values.
- ❖ **Quality Control** – Monitors manufacturing consistency using sample means.
- ❖ **Election Polling** – Predicts voting outcomes from small survey samples.
- ❖ **Financial Market Analysis** – Models stock returns and investment risks.
- ❖ **Machine Learning** – Used in resampling, bootstrapping, and feature scaling.
- ❖ **Insurance & Risk Management** – Predicts claim amounts and sets premiums.
- ❖ **Medical Research** – Generalizes clinical trial results for larger populations.
- ❖ **Supply Chain Management** – Forecasts demand and optimizes inventory.
- ❖ **Fraud Detection** – Identifies anomalies in financial transactions

All of us do not have equal talent. But, all of us have an equal opportunity to develop our talents.

A. P. J. Abdul Kalam




Thank
You!